

Using Image Analysis to Predict Football (Soccer) Penalty Kick Direction

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Abstract

Penalty kicks are crucial in football games. They serve as a tiebreaker for games and as a consequence for teams that foul their opponents in the former's penalty box (Pereira and Patching, 2021). For players, it is a kick that they should score. For goalkeepers, a penalty save could change the tide of the game. Previous research are commonly limited by a low sample data size and testing with non-professional players in non-game-like environments. This project improves on this, collecting 200 penalty kick clips from professional football tournaments spanning the past decade. Snapshots were taken at the time the player kicked the ball in each of the clips. In the snapshot, five features of the players were analyzed, including the non-kicking foot angle, the kicking foot angle, the chest angle, the face angle, and the kicking knee angle. At the time the ball reached the goal net, the distance the ball was from the center of the goal frame was recorded. The positive distance was defined to be to the right of the center and the negative distance as to the left of the center from the player's point of view. Two variables, the non-kicking foot angle and the kicking knee angle, were each found to have a moderately high correlation individually when these variables were compared to the distance the ball ended from the center of the goal. These 2 variables were then combined to create a multi-variable linear equation that could predict the ball distance from the center of the goal. A probability value, 0-1, of the kicker kicking the ball to the right was calculated for each data point based on the multi-variable equation. This probability was used to create a binary value, signaling left and right direction with 0 and 1 respectively. Then, based on the actual direction the ball traveled, a log loss was calculated for each data point. This would penalize more for data points with falsely predicted directions that had a probability value closer to 1 for a predicted right direction and closer to 0 for a predicted left direction. Next, the log loss was minimized to find the constant values in the multi-variable equation. The accuracy value for the finalized predictive model was calculated to be 85 percent accurate in predicting the left or right direction. This study advances information including a highly accurate predictive model that previous research have not included. The study provides information that goalkeepers can use to better predict penalty kicks, which can change the tide for entire games, tournaments, and potentially the team's legacy.

1. Introduction

A standard football goal has a front frame of 7.32 meters in width and 2.44 meters in height and is located at the two ends of the field (“Soccer (Football) Field Templates”). No player may touch the ball with their hands besides the goalkeeper (1 on each team), who can do it only inside their penalty box. The penalty box is a 40.3 m x 16.5 m box surrounding the goal (“Soccer (Football) Field Templates”). Figure 1 shows a diagram of a football pitch with the penalty box and goals labeled.

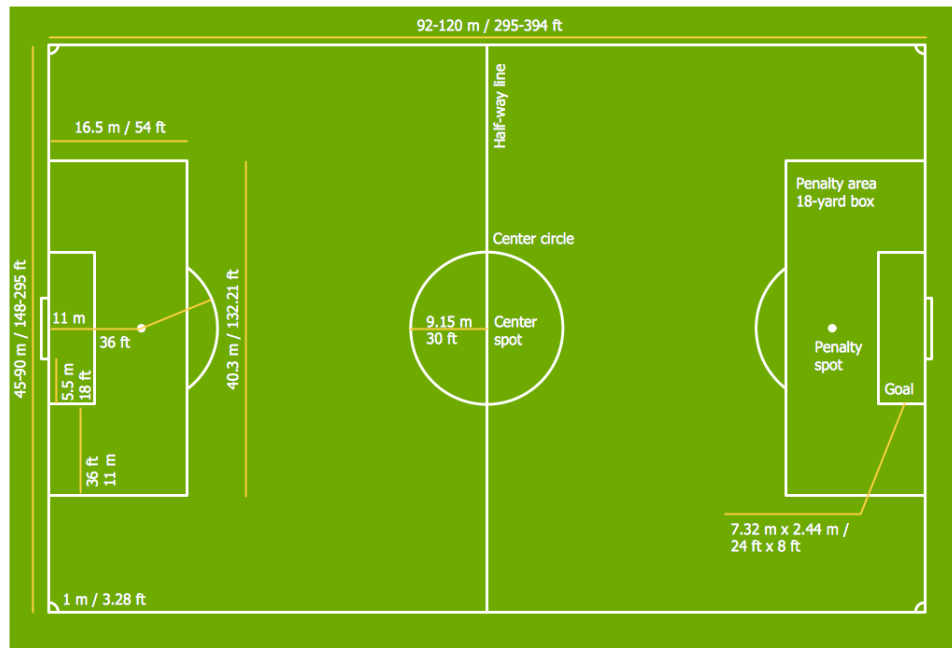


Figure 1: Diagram of football pitch with dimensions labeled including the penalty box area and penalty spot (“Soccer (Football) Field Templates”)

A standard football game consists of two 45-minute halves with up to 30 minutes of extra time during tournaments (Harper et al., 2014). At the end of the game, the team that has scored more goals will win. Generally, there are two ways to score goals: One through open play and the other through set kicks. Set kick examples include penalty kicks, corner kicks, goal kicks, etc.

This research project focuses on penalty kicks. Penalty kicks occur in football for two reasons: One, to act as a tiebreaker for football games that are tied after the standard 90 minutes of regulation time, and two, to act as a consequence for teams that foul their opponents in the former’s penalty box (Pereira and Patching, 2021). During a penalty kick, the kicker shoots a stationary ball positioned 10.97 m away from the center of the goal frame (Pereira and Patching, 2021). Figure 1 depicts a labeled penalty spot dot, showing where the stationary ball will be set to kick. The goalkeeper must stay on the line within the two goalposts until the ball is kicked by the player, which frequently prompts the goalkeeper to dive in one of two directions (left or right). Penalty kick saves by the goalkeeper are critical in football games. These saves can shift the

momentum of a game by boosting confidence for the goalkeeper's team while casting doubt onto the other team (Noël et al., 2014). Ultimately, they have the ability to decide entire games, tournaments, and even a team's lasting legacy. For example: In the most recent FIFA World Cup, the most prestigious tournament in football, the finals between Argentina and France was decided on penalty kicks to break the tiebreaker.

While previous research has studied factors that may help predict penalty kick direction, many contain data from non-professional players with a low sample size (Pereira and Patching, 2021). This study improves on this as its results are based on 200 clips of penalty clips from the most prestigious tournaments (World Cup, Euros, Copa America, Champions League, FA Cup) from the past decade, consisting of some of the best to ever play the sport. The 200 penalty clips collected had a 74.5 percent success rate, closely aligning to the 75 percent success rate of overall penalty kicks (Kiko, 2024). This further verifies the validity of any new information extracted from the study.

Additionally, many studies do not generate research based on game-like environments, where factors such as feelings of pressure, fan chants, etc. could impact the penalty kick. This study improves on this, as data is generated from real penalty kicks from past professional games, where any results concluded can be implemented for real penalty kick scenarios.

Many research studies also fail to create a model that can establish causation and accurately predict the direction of penalty kicks (Li et al., 2015). Instead, they rely their results on the Pearson correlation coefficient. While the Pearson correlation coefficient does imply correlation between two variables, it does not imply causation. In contrast, this project applies a logistic regression model based on the most significant variables found from the 5 variables analyzed (Non-kicking foot angle, Kicking foot angle, Chest angle, Face angle, Kicking knee angle) to generate a prediction model.

The ball's distance from the center of the goal is defined such that a positive distance implies the distance to the right side of the penalty kick kicker and a negative distance implies the distance to the left side of penalty kick taker. A multi-variable equation containing the most significant variables was developed to predict the distance the ball would end up from the center of the goal. From this value, a probability value from 0-1 is calculated for each penalty clip, defined to be the chance the ball is kicked to the right direction in the penalty taker's point of view. A log loss value was then calculated for each data point and the average log loss across the penalty clips was minimized by changing the constant variables in the multi-variable equation. Based on a threshold value, the probability value was categorized to a binary outcome (left or right) and compared to the binary outcome based on the actual direction the ball went (0 = left direction and 1 = right direction from the player's point of view). The threshold value to generate a binary number for the probability value was changed to maximize the amount of clips in which the binary values of the predicted and actual penalty kick direction were equal. This effectively establishes a causation and predictive model for calculating penalty kick direction instead of just proving correlation.

In conclusion, this research project aims to use image analysis to elevate goalkeeper training in saving penalty kicks. Football clips will be analyzed to identify what factors can most effectively predict

penalty kick direction, generating variables that goalkeepers can look for in a player's form to maximize their chances of saving the penalty kick. This will not only further knowledge in predicting football penalty kicks in football, but also set up a base model to analyze specific situations in other sports when examining player form: Pitching in baseball, Throwing in American Football, etc.

2. Computational Approach/Methods

2.1 Data

Variables of the penalty kicker's form were measured on pictures from penalty clips at the time of the player kicking the ball. 5 certain characteristics were measured regarding the player's form (non-kicking foot angle, kicking foot angle, chest angle, face angle, and kicking knee angle), compiled as likely predictive variables based on previous research studies (Lopes et al., 2014).

To collect the most reliable data, 200 right-foot football penalty clips were collected from professional tournaments including the FIFA World Cup, UEFA Euros, Copa America, CONCACAF Gold Cup, Africa Cup of Nations, FA Cup, CONMEBOL Libertadores, Carabao Cup, UEFA Europa League, and Community Shield. Most of these tournaments are among the top 10 most prestigious tournaments in the world. This ensured that any conclusion reached could be applied in professional environments of football. Only right-footed penalty clips were analyzed due to the limited number of left-footed penalty clips available.

For each tournament, we found penalty shootouts available for that tournament on YouTube spanning the past 10 years. Critically, penalty clip film perspectives were consistent across tournaments and years of play: The camera was placed directly behind and above the player shooting the ball. Figure 2 shows a simple diagram of what the filming perspective of the penalty kick was recorded from.

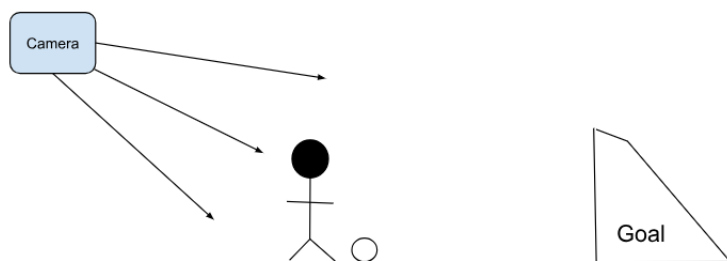


Figure 2: Diagram of where the camera is positioned relative to the penalty taker in the football clips from all of the data analyzed.

2.2 Image Analysis

In this sense, the analysis was conducted on images with the same camera angle, which will prevent inaccuracies of measuring angles due to camera perspective. In analyzing each image, the video was slowed to 0.25x speed and paused at the exact time each player made contact with the ball. Using the snipping

tool, a screenshot capturing the player's form when kicking the ball was taken from the YouTube video.

The software Wolfram Mathematica was used to analyze the images. The first program, angleTool, was created to analyze specific angle characteristics in the form of the football player at the time of kicking the ball. Figure 3 shows a snippet of this code, where 3 interactive points are created that can be moved around in an image to form an angle. The 3 points p1, pt, and p2 are connected by two lines, one from p1 to pt and one from pt to p2. The point pt would then indicate the vertex point where we use to calculate the angle. The VectorAngle method in the last line of the code snippet is used to calculate this angle. The degrees follow the smaller angle formed by the two lines. When measuring angles, one of the lines is kept directly horizontal, and the other line rotated around to measure the angle.

```
Locator[Dynamic[offset + {1, -1}, (offset = # - {1, -1}) &], Appearance -> Graphics[{Red, EdgeForm[Black], Disk[]}]],
Locator[Dynamic[offset + {1, 1}, (offset = # - {1, 1}) &], Appearance -> Graphics[{Blue, EdgeForm[Black], Disk[]}]],
Locator[Dynamic[offset + pt, (pt = # - offset) &], Appearance -> Graphics[{Yellow, EdgeForm[Black], Disk[]}]],
Locator[Dynamic[offset + p1, (p1 = # - offset) &], Appearance -> Graphics[{Yellow, EdgeForm[Black], Disk[]}]],
Locator[Dynamic[offset + p2, (p2 = # - offset) &], Appearance -> Graphics[{Yellow, EdgeForm[Black], Disk[]}]],
Dynamic@Text[Style[Round[VectorAngle[p1 - pt, p2 - pt] * 180 / Pi, 0.1] °, Red, 16], Offset[{20, -20}, offset + pt]], ImageSize -> 300, PlotRange -> 1.5],
```

Figure 3: Snippet of code that creates 3 interactive points that can be moved around to calculate the angle of a specific feature in an image.

The angleTool program was used to label 5 features of each image: Non-kicking foot angle, Kicking foot angle, Chest angle, Face angle, and the Kicking knee angle. After the features of the player form were labeled, the video was resumed at 0.25x speed until the moment the ball reaches the back net of the goal or the gloves of the goalkeeper.

Next, the distanceTool program was coded to analyze the distance that the player shot the ball from the center of the goal. This will be our dependent variable when analyzing the Coefficient of determination later. From the kicker's point of view, the center of the goal was set as 0; any distance to the right indicated a positive distance, and any distance to the left indicated a negative distance. This will be important later when examining and developing binary values to represent penalty kick direction. Figure 4 shows a code snippet of the distanceTool program. In this program, two interactive points pt1 and pt2 are created. The y position of both points are set to be the same to ensure that the distance calculated is purely horizontal. The last line of the code snippet then calculates the distance between the two points. The horizontal length of the image inputted is set to a distance of 1 and the distance measurements are relative to this image length.

```
Deploy@Grid[{{Graphics[{Inset[Image[g, ImageSize -> 300]], Dynamic@Line[{pt1, pt2}], Locator[Dynamic[pt1, (pt1 = {#[[1]], yPos = #[[2]]};
pt2 = {pt2[[1]], yPos}) &], Appearance -> {Graphics[{Blue, Disk[]}], {Automatic, Automatic}}], Locator[Dynamic[pt2, (pt2 = {#[[1]], yPos = #[[2]]};
pt1 = {pt1[[1]], yPos}) &], Appearance -> {Graphics[{Green, Disk[]}], {Automatic, Automatic}}],
Dynamic@Text[Style[N[EuclideanDistance[pt1, pt2], 2], Red, 16], Scaled[{0.8, 0.2}]]}, ImageSize -> 300, PlotRange -> {{0, 1}, {0, 1}},
```

Figure 4: distanceTool code snippet that creates two interactive points that can be moved to calculate a certain distance.

The image inputted was taken to precisely cover the goal frame's horizontal length. Then, the two interactive points from the distanceTool method would measure the ball's distance from the center of the goal once the ball reaches the net. This distance calculated from the program would be converted to the real distance by multiplying it by the football goal length of 7.32 meters.

Figure 5 depicts an example of how the distance was calculated for a penalty clip. In the example, the distanceTool calculates that the distance of the ball from the very right side of the goal frame is 0.768846. To calculate the distance from the center of the goal, we simply subtract 0.5 from 0.768846 as 0.5 will be the width of half the goal frame. When entering this data in Excel, we will enter in -0.268846 because the ball was to the left of the center of the goal frame.



Figure 5: The distanceTool program is used to calculate the distance the ball is from the center of the goal at the time of reaching the net.

Penalty clips in which the ball was kicked outside of the goal frame were excluded from analysis as the ball would never reach the goal net. Moreover, since only right foot penalty clips were analyzed, left-footed penalty clips were removed.

In instances where the goalkeeper saved the kick, the video was paused at the time the ball reached the goalkeeper's gloves or body. Then, a straight edge was used to predict where the ball would end up at the goal net and the distanceTool program used this newly made point to measure a distance from the center of the goal.

Figure 6 shows the general steps of how images in the project were analyzed to label data.

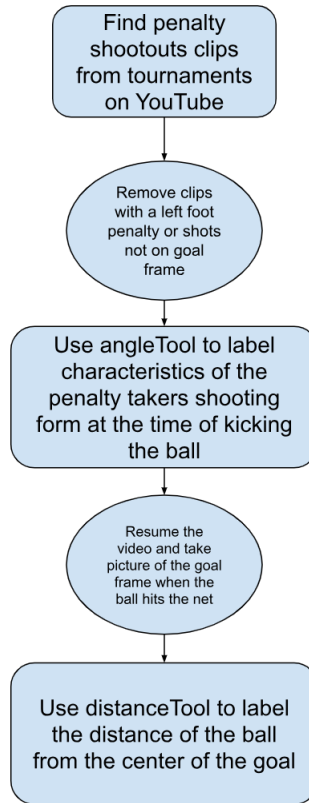


Figure 6: Flow diagram showing process of labeling data for the project

2.3 Analysis of data

We then created separate lines of best fit for the distance from the center of the goal at the time the ball reaches the net vs. each of the 5 variables of the penalty taker's kicking form at the time of kicking the ball (Non-kicking foot angle, Kicking foot angle, Chest angle, Face angle, Kicking knee angle). After labeling these variables of the player form for 76 penalty clips spanning various professional tournaments, two variables (Non-kicking foot angle, Kicking knee angle) showed some significance in their correlation with where the penalty kick ended relative to the center of the goal.

A graph of one of the variables, Distance vs. Non-kicking foot angle, is shown below. The Distance vs. Kicking knee angle had a similar trend, where as the angle increased, the distance of the ball from the center of the goal gradually decreased, eventually changing from positive to negative, or from right to left direction. When compared to the distance from the center of the goal the ball ended up being kicked, both of the variables, non-kicking foot angle and kicking knee angle, had coefficient of determination values of around 0.4.

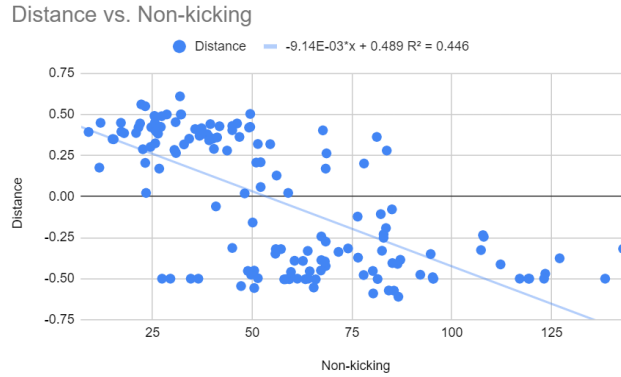


Figure 7: The distance decreases as the non-kicking foot angle increases

Since only the two variables (Non-kicking foot angle, Kicking knee angle) discussed show any significance when compared to the distance of the ball from the center of the goal, only the Non-kicking foot angle and Kicking knee angle were analyzed for each penalty clip relating to the kicker's form moving forwards, until 200 right-foot penalty clips were labeled.

A systematic selection system was used on the 200 penalty clip's variable labeling information, such that labeling data from every alternating penalty clip was used as the training data, resulting in 100 training data points. The other 100 penalty clip will be used as testing data later on.

Using the training data, the labeling data from the 100 penalty clips were entered into an Excel spreadsheet.

We decided to implement a logistic regression model in creating a prediction model that can predict between binary outcomes.

We first set up a multi-variable equation in the format of:

$$Ax + By + C = Z$$

Figure 8: Formula to calculate predicted value Z

In the equation: A, B, and C are designated constants that were initially set to 0.5 while x and y represent the non-kicking foot angle and kicking knee angle respectively, with Z representing the distance the ball is predicted to end up from the center of the goal at the time of hitting the net. For Z, a positive distance represents the distance the ball ended up to the right of the center of the goal frame from the kicker's point of view, while a negative distance represents the distance the ball ended up to the left of the center of the goal frame from the kicker's point of view.

Based on the predicted value, Z, from our initially set multi-variable equation, a probability value was calculated for each Z value. This represented the probability that the ball would in the right direction

from the kicker's point of view. This probability value was calculated following the logistic function for logistic regression models:

$$e^Z / (1 + e^Z) = P$$

Figure 9: Formula to calculate probability that the ball goes to the right based on the predicted value Z from figure 8

Z represents the predicted value calculated in the formula from figure 8 and P represents the probability that the ball is kicked to the right direction from the penalty taker's point of view. In other words, right is defined to be any positive value for the distance from the center of the goal.

To develop binary values, a value B1 was assigned to each probability value such that any value 0.5 (threshold value) or greater would be assigned a 1 and any value smaller than 0.5 would be assigned a 0.

The actual distance from the center of the goal at the time the ball reached the net extracted from the training data was also used to calculate a binary outcome (B2), such that $B2 = 1$ for any value (distance from center of goal) 0 and greater, and $B2 = 0$ for any value (distance from center of goal) less than 0.

Using the B2 value, a log loss value was calculated for each training penalty clip labeled. The log loss was computed such that if B2 was equal to 1, then the log loss for the particular data point would equal to $-\text{Log}(P)$ for that data point. If B2 was equal to 0, then the log loss for the particular data point would be calculated using the equation $-\text{Log}(1 - P)$. In this way, data points in which the ball was kicked opposite the predicted distance would be penalized more (greater log loss value) the "higher" the probability value for its initial predicted direction. A "higher" probability value for a predicted left direction from the kicker's point of view would be a P value closer to 0, while a higher probability value for a predicted right direction from the kicker's point of view would be a P value closer to 1.

Using the Solver add in tool on Excel, we set the parameters to minimize the average log loss value across the training data by changing the values A, B, and C of the multi-variable equation. Figure 10 below shows how the values were inputted into the Solver-add in function in Excel.

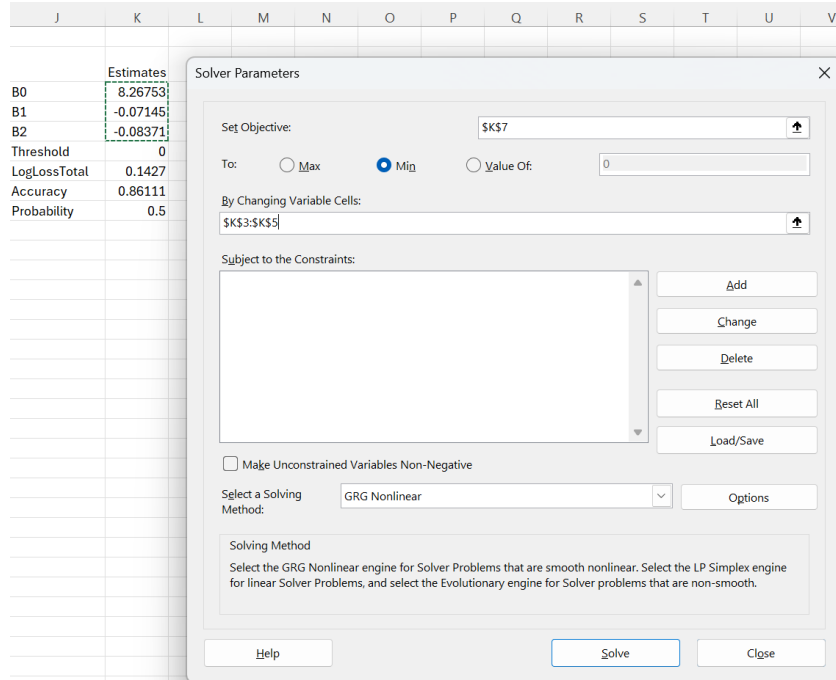


Figure 10: The Solver tool was used to change the values A, B, and C of the equation in figure 8 to minimize the average log loss across all data points

With the newly created multi-variable equation, the B1 value for each training data was altered following the same guidelines set earlier in creating binary values.

At this stage, we can now calculate an accuracy value for the newly created multi-variable equation in predicting the binary classified directions (left and right) of penalty kick direction.

By comparing B1 (calculated from the probability, P, value) and B2 (the actual direction the ball traveled in), an accuracy value was calculated by counting every data point in which $B1 = B2$ as accurate and all other data points inaccurate.

The Solver tool was used again to maximize the accuracy value by changing the threshold value (originally set to 0.5), which was used as the cutoff P value to determine the binary value, B1.

2.4 Testing the data

The multi-variable predictive equation developed was then tested on the testing data, the other 100 penalty clips labeled, and the accuracy values were compared to determine the validity of the predictive equation.

3. Results

To analyze the correlation between each specific variable to the distance the ball ended up from the center of the goal, the Coefficient of determination was calculated using the equation:

$$r^2 = \frac{\Sigma(a-b)^2}{\Sigma(b-a)^2}$$

Figure 11: Formula to calculate the Coefficient of Determination r^2

such that the value a represents the actual ball distance from the center of the goal and the value b represents the predicted distance based on the best-fit line of the variable vs. distance graph. The Coefficient of Determination for each variable (non-kicking foot angle, kicking foot angle, chest angle, face angle, kicking knee angle) was first calculated for the 76 penalty discussed earlier. Figure 12 illustrates a scatter plot of each variable (in degrees) when compared to the distance of the ball from the center of the goal (in meters).

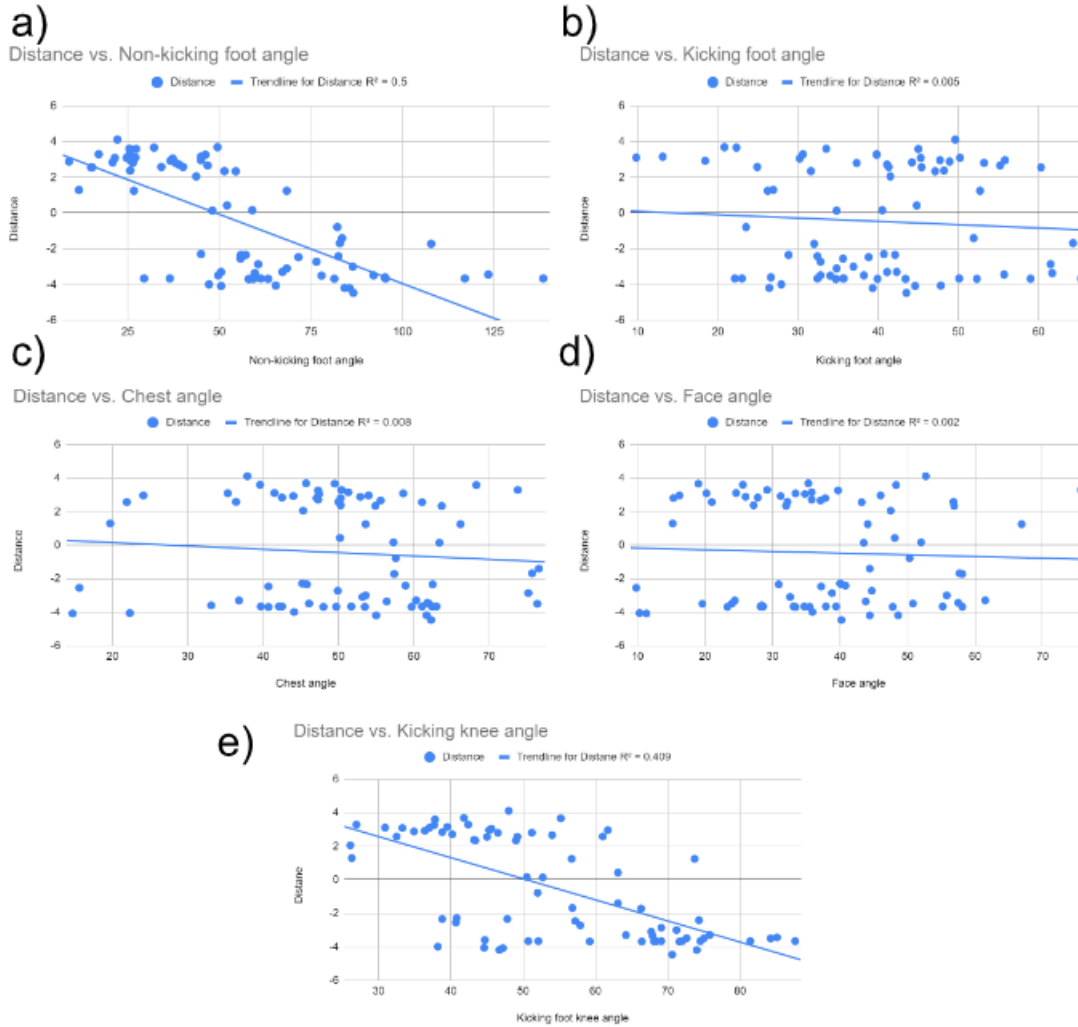


Figure 12: Coefficient of determination values were calculated for the best fit lines of each variable compared with the penalty kick's distance from the center.

Two of the five variables, the Non-kicking foot angle and the Kicking knee angle had moderate to high Coefficient of determination values of 0.5 and 0.409 respectively. This means that around 50 percent of the variance in the ball's distance from the center of the goal can be explained by the Non-kicking foot angle. Likewise, 40.9 percent of the variance in the ball's distance from the center of the goal can be explained by the Kicking knee angle. The other 3 variables (Kicking foot angle, Chest angle, Face angle) all had Coefficient of determination values less than 0.01, meaning that their correlation with the penalty kick direction value was insignificant.

Thus, when moving forward, we only labeled each penalty kick clip with 2 variables: Non-kicking foot angle and the Kicking knee angle. At the end of labeling 200 penalty kick clips, Figure 13 shows the scatter plot and Coefficient of determination values for the Non-kicking foot angle and Kicking knee angle.

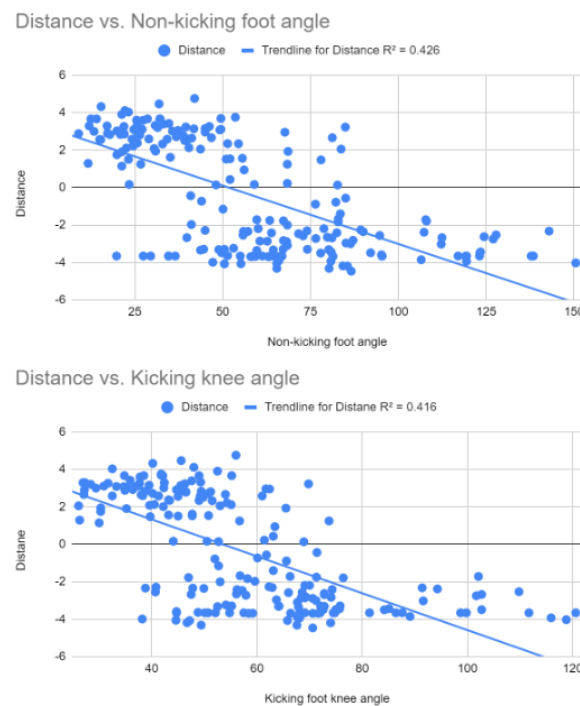


Figure 13: Scatter plot of the Non-kicking foot angle and the Kicking knee angle, including the line of best fit and Coefficient of determination values

Similar to the values of Figure 12, both Coefficient of determination values for the Non-kicking foot angle and the Kicking knee angle when compared to the distance the ball ended up from the center of the goal were between 0.4 and 0.5 with values of 0.426 and 0.416 respectively. This means they have a moderate to high value of correlation.

To make a prediction model, these 2 variables were then used in a multi-variable equation, represented by:

$$Ax + By + C = Z$$

Figure 14: Formula to calculated predicted value Z

Using the 100 penalty clips that were systematically selected from the 200 total right-foot penalty clips labeled, a logistic regression was then ran with the process described in 2.3: Analysis of data on the training set.

Figure 15 shows the probability of the penalty kick direction going to the right direction from the kicker's point of view when compared to the Non-kicking foot angle and the Kicking knee angle respectively. These graphs included all of the training data points. It is clear that the probability value decreases as the angles of both variables increase.

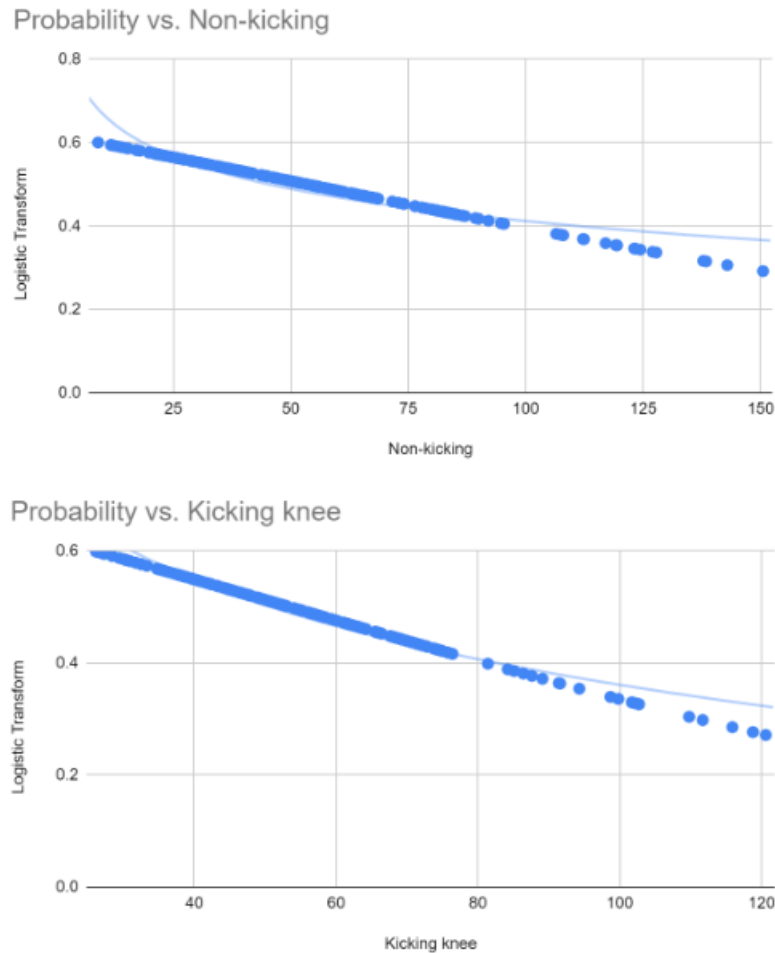


Figure 15: Graph representing the probability of the penalty direction going to the right from the kicker's point of view compared to the Non-kicking foot angle (top) and Kicking knee angle (bottom)

Using the multi-variable equation with two constant parameter weights initially set at 0.5, the log loss was calculated as described in 2.3. Analysis of data. The average log loss value across the testing data set was then minimized to 0.1427 by changing the constant values using the Solver Add-in tool, resulting in $A = 8.26753$, $B = -0.07145$, and $C = -0.08371$. These constant variables were substituted back into the multi-variable equation in Figure 8 to re-calculate the binary values B1.

The multi-variable equation's model can be modeled with the 3-D plot shown in Figure 16. It can be recognized that as the Non-kicking foot angle and the Kicking knee angle increases, the plane's height, representing the distance from the center of the goal begins to drop below 0, indicating a penalty kick shifting towards the left direction. Similarly, as the Non-kicking foot angle and Kicking knee angle decreases, the plane in Figure 16 begins to rise above the XY plane, representing a positive ball distance from the goal, or the right direction.

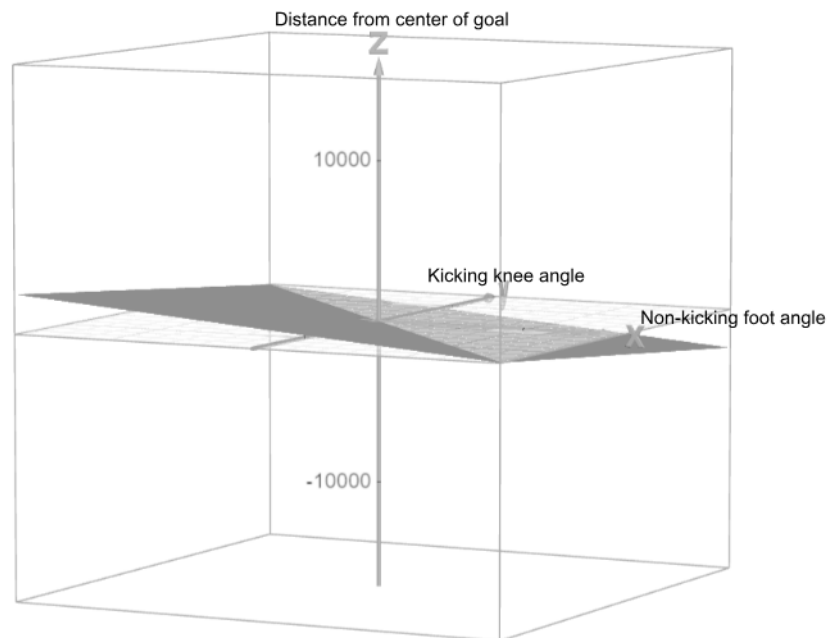


Figure 16: A 3-D plot representing the multi-variable equation involving the Non-kicking foot angle and the Kicking knee angle.

Ultimately, by calculating the rate when $B1 = B2$, we reached an 85 percent accuracy rate on the training data set. This newly made multi-variable equation was then tested on the other half of the 200 penalty clips, the testing data set, resulting in an accuracy rate that was also 85 percent. This means that the predictive equation is valid as its accuracy rate is consistent when used across the training and testing data. Compared to the otherwise 50 percent chance when guessing randomly, this model's accuracy in predicting penalty kicks is significant. Using information from the model, goalkeepers can better predict and save penalty clips in the future by looking at the angles of the player's kicking foot angle and kicking knee angle when they are kicking the penalty kick to anticipate the direction of the penalty kick.

4. Discussion and Conclusion

This study developed a prediction model that can be used to predict football penalty kicks by looking at certain characteristics of the player when kicking the penalty kick. Unlike any previous research, the study uses a significant amount of penalty kick clips from premier tournaments spanning the past decade: FIFA World Cup, UEFA Euros, Copa America, Fa Cup, Africa Cup of Nations, etc. This improves on past research that experimented with unprofessional players not playing at the sport's highest level, which likely made any of the results extracted unreliable for professional environments.

By using the non-kicking foot angle and kicking knee angle as variables in the equation, a prediction model was created with a 85 percent binary accuracy rate when predicting left or right direction. The testing data resulted in the exact same accuracy rate (85 percent), verifying that the predictive equation's accuracy rate is consistent across all data sets.

There are many ways to use information extracted from the study. Trainers can use this information to elevate goalkeeper training, running practice penalty kicks to train the goalkeeper's split decision instinct to dive in one of two directions based on the kicker's non-kicking foot and kicking knee angle. On the other side, penalty kick takers can also use this information to learn ways to better hide what way they are kicking by positioning their non-kicking foot angle and kicking knee angle to not follow the general trend found in the study.

4.1 Future

In the future, the information extracted from this study will be tested on real goalkeepers across the nation. 100 penalty clips will be selected randomly across the 200 clips analyzed to show to goalkeepers. Each clip will be paused at the time the kicker kicks the ball, and the goalkeepers will be prompted to predict the left or right direction that the ball will go. Then, the clip will be resumed and show the actual direction the ball went. Across all 100 clips, an accuracy of the goalkeeper's prediction will be calculated.

Next, the goalkeeper will be shown the predictive model that we developed, showing what variables to look for and how it will impact the probability that the player will kick left or right. After analyzing the model, the goalkeeper will now be tested on the other 100 clips and their new prediction accuracy rate will be calculated. If the new prediction accuracy rate increases a significant amount from the goalkeeper's original prediction accuracy rate, then this can further validate the prediction model's power in helping goalkeepers predict penalty kicks effectively.

Furthermore, the data size of this project can be expanded to include professional clips spanning more tournaments. To create better data, professional football players could be analyzed in a live setting too, where markers can be adjusted onto different parts of the player's clothes to measure variable angles, ensuring more accurate data labeling. Similarly, the research's prediction model could also be tested in a live setting in which the goalkeepers have to make a decision in real time instead of through a screen, which could further validate the model.

Player forms across other sports can also be applied using the base model of this project. For

instance, a predictive model could also be applied to the baseball's pitchers throw direction based on the pitcher's form, or where a quarterback in American football is looking to throw the football based on the quarterback's form.

Relating to this research, a device can also be created to analyze a player's form - whether in football, baseball, or tennis - and provide feedback to optimize a player's form to create the most effective results in the sport. For example, the device could optimize a player's form in football to kick the most effective shots for free kicks by providing the player feedback on how much to lean over the ball, where to place the non-kicking foot, etc.

References

- ConceptDraw. "Soccer (Football) Field Templates."
- Harper, Liam D., et al. "Technical performance reduces during the extra-time period of professional soccer match-play." *PLOS ONE*, vol. 9, no. 10, 2014, pp. 1–6. <https://doi.org/10.1371/JOURNAL.PONE.0110995>.
- Kiko. "Interesting Penalty Kick Statistics You Must Know in 2024." Feb. 2024.
- Li, Yumeng, et al. "Prediction of Kick Direction from Kinematics during the Soccer Penalty Kick." *International Journal of Kinesiology and Sports Science*, vol. 3, Oct. 2015, pp. 1–7. <https://doi.org/10.7575/aiac.ijkss.v.3n.4p.1>.
- Lopes, José E., et al. "Predicting the lateral direction of deceptive and non-deceptive penalty kicks in football from the kinematics of the kicker." *Human Movement Science*, vol. 36, 2014, Including Special Section articles of Developmental Coordination Disorder and Motor Control (DCD- IX International Conference - Lausanne 2011), pp. 199–216. <https://doi.org/https://doi.org/10.1016/j.humov.2014.04.004>.
- Noël, Benjamin, et al. "The development of a method for identifying penalty kick strategies in association football." *Journal of Sports Sciences*, vol. 33, no. 1, 2014, pp. 1–10. <https://doi.org/10.1080/02640414.2014.926383>.
- Pereira, Mauro R., and Geoffrey R. Patching. "Goal Side Selection of Penalty Shots in Soccer: A Laboratory Study and Analyses of Men's World Cup Shoot-Outs." *Perceptual and Motor Skills*, vol. 128, no. 5, Oct. 2021, pp. 2279–303. <https://doi.org/10.1177/00315125211025412>.